

Precept 2 Outline

ECO 202 TAs

September 17, 2026

1 Weekly check-in

1. How are things going? Questions or comments?

2 Review of practice questions

1. **(Practice 3, Question 2)** Two historical occupational groups have the following sample summaries for annual earnings, measured in thousands of dollars:

Group	Sample mean	Sample standard deviation	One worker's earnings
A	60	8	72
B	75	10	90

- (a) Calculate the sample z-score of the listed worker in each group.

Solution. $z_A = (72 - 60)/8 = 1.5$ and $z_B = (90 - 75)/10 = 1.5$. Thus $z_A = z_B = 1.5$.

- (b) One worker earns more in raw dollars. Who has the higher standardized position within the corresponding group? Explain the difference between the two comparisons.

Solution. The Group B worker earns more: $90 > 72$. Both are 1.5 standard deviations above their group means. Thus B earns more; their standardized positions are equal.

- (c) In Group A, what earnings correspond to $z = -0.75$? Reverse the standardization and state the answer in thousands of dollars and in dollars.

Solution. $x = \bar{x} + zs = 60 + (-0.75)(8) = 54$. Thus 54 thousand dollars = \$54,000.

- (d) Show that expressing every Group A amount in dollars rather than thousands of dollars leaves the z-score unchanged.

Solution. In dollars, $z = (72,000 - 60,000)/8,000 = 12,000/8,000 = 1.5$. Multiplying all amounts by 1,000 cancels from the ratio.

- (e) Does $z = 1.5$ by itself establish an exact percentile or a causal explanation for earnings? State what additional model or design information would be needed for either claim.

Solution. A percentile requires the group's distribution; under a Normal model, $\Phi(1.5) = 0.9332$. A causal claim requires random assignment or another credible identification design and its assumptions. Thus $z = 1.5$ alone implies neither.

2. **(Practice 3, Question 4)** In a newly constructed teaching example, weekly hours for a group of full-time workers are approximated by

$$X \sim N(40, 4^2)$$

You may use the following standard Normal values:

- (a) Identify the model mean, standard deviation, and variance, with units.

Solution. $\mu = 40$ hours, $\sigma = 4$ hours, $\sigma^2 = 16$ hours².

Standard Normal quantity	Value
$\Phi(-1.5)$	0.0668
$\Phi(-1)$	0.1587
$\Phi(1)$	0.8413
$\Phi(2)$	0.9772
90th-percentile z -value	1.2816

- (b) Calculate and interpret $P(X < 34)$.

Solution. $z = (34 - 40)/4 = -1.5$, so $P(X < 34) = \Phi(-1.5) = \boxed{0.0668}$. About 6.68% work fewer than 34 hours.

- (c) Calculate and interpret $P(36 < X < 44)$.

Solution. The standardized limits are -1 and 1 , so $P(36 < X < 44) = \Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = \boxed{0.6826}$.

- (d) Calculate and interpret $P(X > 48)$ using a complement.

Solution. $z = (48 - 40)/4 = 2$, so $P(X > 48) = 1 - \Phi(2) = 1 - 0.9772 = \boxed{0.0228}$.

- (e) Find and interpret the model's 90th percentile in hours.

Solution. $x_{0.90} = 40 + (1.2816)(4) = 45.1264$, so $x_{0.90} \approx 45.13$ hours. About 90% work fewer hours.

- (f) Actual weekly work hours can be heaped at round numbers and may not be symmetric. Explain why the answers above are model-based approximations and name a graphical check relevant to this question.

Solution. The calculations assume the actual distribution is well approximated by a continuous, symmetric Normal curve. Check a histogram or Normal Q-Q plot, especially for heaping and skewness.

3. **(Previous ECO 202 HW Question)** Consider the distribution of X characterized by the density curve

$$f(x) = \begin{cases} \frac{1}{2} & \text{for } 0 \leq x \leq 1 \\ \frac{3}{4} - \frac{1}{4}x & \text{for } 1 < x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Draw the density curve and check that the area below the curve is one.

Solution. The graph is horizontal at $1/2$ on $[0, 1]$, then linear from $(1, 1/2)$ to $(3, 0)$. Its area is $1(1/2) + \frac{1}{2}(2)(1/2) = \boxed{1}$.

- (b) Compute the median of this distribution.

Solution. The area on $[0, 1]$ is $1(1/2) = 1/2$, so $\text{Median}(X) = 1$.

- (c) What fraction of observations of X are larger than 2?

Solution. The region from 2 to 3 is a triangle with base 1 and height $f(2) = 1/4$, so $P(X > 2) = \frac{1}{2}(1)(1/4) = \boxed{\frac{1}{8}}$.

- (d) **(Challenge)** Provide the density curve that characterizes the distribution of $Y = X/2$.

Solution. Since $X = 2Y$, $f_Y(y) = 2f_X(2y)$. Thus $f_Y(y) = 1$ on $[0, \frac{1}{2}]$; $f_Y(y) = \frac{3}{2} - y$ on $(\frac{1}{2}, \frac{3}{2}]$; 0 otherwise.

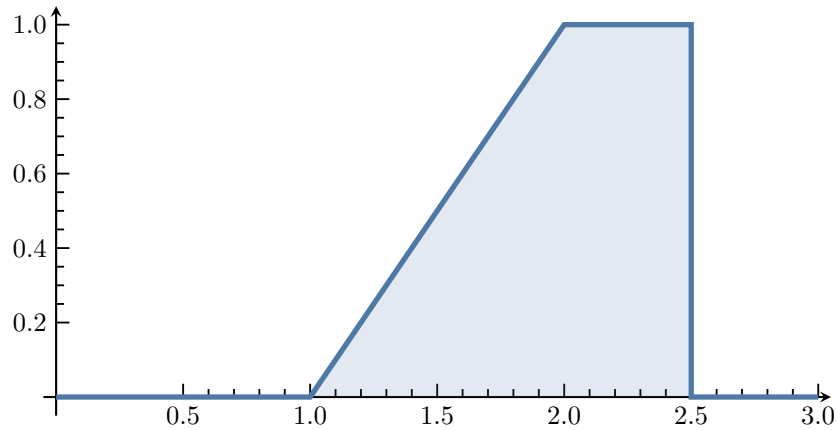
3 Extra practice problems

Note: Some of these are challenging!

1. Let X be the distribution described by the following density curve:

- (a) What is the median of X ?

Solution. The triangle on $[1, 2]$ has area $\frac{1}{2}(1)(1) = 1/2$. Hence $P(X \leq 2) = 1/2$ and $\text{Median}(X) = 2$.



- (b) Is the mean smaller or larger than the median? Explain.

Solution. The triangle and rectangle each have area $1/2$, with centroids $5/3$ and $9/4$. Thus $E[X] = \frac{1}{2}(5/3) + \frac{1}{2}(9/4) = 47/24 \approx 1.958 < 2$, so the mean is smaller.

- (c) What fraction of $(X + 2)/7$ is smaller than $1/2$?

Solution. $(X + 2)/7 < 1/2$ iff $X < 3/2$. The area from 1 to $3/2$ is a triangle: $P(X < 3/2) = \frac{1}{2}(\frac{1}{2})(\frac{1}{2}) = \boxed{\frac{1}{8}}$.

2. The distribution of the SAT scores of applicants to a college is approximately distributed normal with mean 1280 and standard deviation 100.

- (a) What fraction of applicants have an SAT score between 1350–1450?

Solution. The z -scores are $(1350 - 1280)/100 = 0.7$ and $(1450 - 1280)/100 = 1.7$. Thus $P(1350 < X < 1450) = \Phi(1.7) - \Phi(0.7) = 0.9554 - 0.7580 = \boxed{0.1974}$.

3. Suppose the weight (in carat) of a collection of 456 gemstones is approximately normal with mean 2.1 and standard deviation 0.7.

- (a) Approximately how many gemstones weigh more than 2.4 carats?

Solution. $z = (2.4 - 2.1)/0.7 \approx 0.43$, so $456[1 - \Phi(0.43)] \approx 456(0.334) = 152.3$. Thus about 152 gemstones.

- (b) Suppose the gemstones are sorted by weight. What is the approximate weight of the 175th heaviest gemstone?

Solution. The fraction below it is approximately $1 - 175/456 = 0.6162$, whose z -value is about 0.30. Thus $x \approx 2.1 + (0.30)(0.7) = \boxed{2.31 \text{ carats}}$.

4. A quantitative variable x is observed for $n = 6$ individuals and takes on the values 2, 4, 5, 7, 8, 13.

- (a) Sketch the boxplot of this distribution. Make sure you put numbers on the y-axis.

Solution. $Q_1 = 4$, median $= (5 + 7)/2 = 6$, and $Q_3 = 8$. The box runs from 4 to 8, with median 6 and whiskers at 2 and 13: (2, 4, 6, 8, 13).

- (b) Is 13 an outlier according to the rule of thumb? Explain.

Solution. $IQR = 8 - 4 = 4$, so the upper fence is $8 + 1.5(4) = 14$. Since $13 < 14$, 13 is not an outlier.

- (c) Find another distribution of $n = 6$ values with the same median and interquartile range as x , but with mean equal to 20.

Solution. Keep the first five values and choose the sixth so the sum is $6(20) = 120$: $120 - (2 + 4 + 5 + 7 + 8) = 94$. Thus 2, 4, 5, 7, 8, 94, with median 6 and $IQR = 8 - 4 = 4$.

5. The quantitative variable x is approximately normally distributed with mean 10 and standard deviation 3.

- (a) What fraction of values of x are approximately larger than 12?

Solution. $z = (12 - 10)/3 = 2/3$, so $P(X > 12) = 1 - \Phi(2/3) \approx \boxed{0.2525}$.

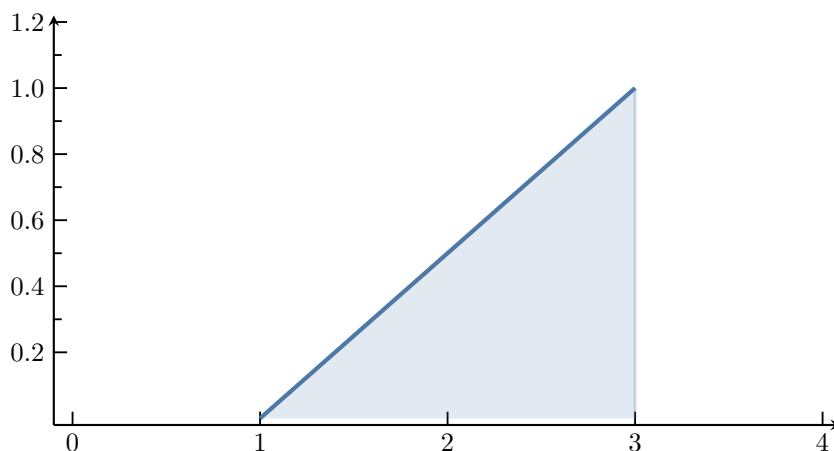
- (b) What values of x smaller than 10 are so small that they would be considered outliers according to the rule of thumb for outliers?

Solution. For this Normal distribution, $Q_1 = 10 - 0.6745(3) = 7.9765$, $Q_3 = 12.0235$, and $IQR = 4.0469$. The lower fence is $7.9765 - 1.5(4.0469) \approx 1.906$, so $\boxed{x < 1.91}$.

- (c) Find a normal distribution with variance smaller than 10 such that the fraction of values that are approximately larger than 12 is the same as that of x .

Solution. The standardized cutoff must remain $2/3$. Choose $\sigma = 2$ and solve $(12 - \mu)/2 = 2/3$, giving $\mu = 32/3$. One answer is $\boxed{N(32/3, 2^2)}$, whose variance is $4 < 10$.

6. The graph below depicts the density curve of the distribution of some quantitative variable x .



- (a) What fraction of the values of x is larger than 2?

Solution. The area below 2 is a triangle with base 1 and height $1/2$: $P(X \leq 2) = \frac{1}{2}(1)(1/2) = 1/4$. Thus $P(X > 2) = \boxed{3/4}$.

- (b) Sketch, on the same graph, the density curves of x , $y = x - 2$ and $w = 2x$. Put numbers on both axes.

Solution. The line segments defining the positive-density regions are $X : (1, 0) \rightarrow (3, 1)$; $Y : (-1, 0) \rightarrow (1, 1)$; $W : (2, 0) \rightarrow (6, 1/2)$. For W , density heights are divided by 2 so its area remains one.

7. Suppose the distribution of some quantitative variable x is approximately normal with mean μ and standard deviation 4.

- (a) Suppose $\mu = 1$. What approximate fraction of x is larger than 3?

Solution. $z = (3 - 1)/4 = 0.5$, so $P(X > 3) = 1 - \Phi(0.5) = 1 - 0.6915 = \boxed{0.3085}$.

- (b) Without assuming $\mu = 1$, consider the approximate fraction of x larger than 3. For what values of μ is this fraction smaller than 0.75? Explain.

Solution. We need $1 - \Phi((3 - \mu)/4) < 0.75$, so $(3 - \mu)/4 > \Phi^{-1}(0.25) \approx -0.6745$. Therefore $\boxed{\mu < 3 + 4(0.6745) \approx 5.70}$.